MSc Data Science and Computational Intelligence

Individual Project Preparation Module

Research Proposal

Design and Development of a

Football Betting Prediction Model

Using Statistical Inference and Machine Learning

Evidence from Europe's Top Five Football Leagues

Premier League • La Liga • Serie A • Bundesliga • Ligue 1

Submitted by:

[Student Name]

Student ID: [ID Number]

Supervisor:

[Supervisor Name]

December 2025

Keywords

MACHINE LEARNING Football PREDICTION

Statistical Inference BETTING MODEL

Probability DATA ANALYSIS Regression

PREMIER LEAGUE Classification ODDS

La Liga RANDOM FOREST Serie A

Expected Goals LOGISTIC Bundesliga

POISSON DISTRIBUTION Ligue 1

Feature Engineering VALUE BETTING

Python ACCURACY Validation

Figure 1: Visual representation of key concepts
(Size indicates relative importance in the research)

Contents

Table of Figures

Table of Tables

Introduction

Football is widely regarded as the most popular sport in the world. It is watched by billions of people across different continents and cultures. The game has evolved significantly over the past decades, not only in terms of how it is played but also in how it is analysed and understood. Modern football is no longer just about talent and tactics on the pitch. It has become a data-driven enterprise where numbers and statistics are used to gain competitive advantages.

The betting industry has grown alongside football's popularity. It is estimated that billions of pounds are wagered on football matches every year across Europe alone. This growth has been fuelled by the availability of live data, online platforms, and increasingly sophisticated odds-setting mechanisms employed by bookmakers. However, despite the vast amounts of money involved, most bettors consistently lose money over time. This is because bookmakers have access to advanced statistical models and historical data that the average punter does not possess.

This research proposal outlines a project that seeks to explore whether a data-driven betting model can be developed using publicly available data from Europe's top five football leagues. These leagues are the English Premier League, Spanish La Liga, Italian Serie A, German Bundesliga, and French Ligue 1. The project will apply statistical inference techniques and machine learning algorithms to historical match data spanning over three decades from 1990 to 2026. The aim is not to create a perfect prediction system but rather to understand the underlying patterns in football match outcomes and to develop a deeper appreciation of the challenges involved in sports prediction.

Figure 1: Data Coverage Timeline (1990-2026)

36 YEARS OF FOOTBALL DATA

1990

2026

~70,000 Matches • 5 Leagues • 36 Seasons

🏴 Premier League ES La Liga IT Serie A DE Bundesliga FR Ligue 1

Figure 2: Evolution of Football Analytics (1990-2026)

ERA 1: 1990-2005	ERA 2: 2005-2015	ERA 3: 2015-2026
Traditional Statistics Goals • Assists • Cards	Advanced Metrics Possession • Pass Accuracy	Data Science Era xG • ML • Tracking Data

(Dataset captures football's analytical transformation across three distinct eras)

Motivation

The motivation for this project stems from a genuine interest in both football and data science. As someone who has followed football for many years, there has always been curiosity about what makes certain teams win consistently while others struggle. Traditional explanations such as squad quality, managerial ability, and home advantage provide some answers. However, these explanations are often subjective and difficult to quantify.

Data science offers an opportunity to investigate these questions more rigorously. By collecting and analysing historical match data, it becomes possible to identify patterns that might not be obvious to the casual observer. For example, certain teams might perform better in specific conditions, or particular statistical indicators might be more predictive of match outcomes than others. Understanding these patterns has both academic and practical value.

Furthermore, the betting market provides an interesting test case for prediction models. Bookmaker odds are essentially probability estimates set by experts using sophisticated methods. If a model can consistently identify situations where these odds appear to be mispriced, it suggests that the model has captured something meaningful about football match dynamics. This does not mean that profitable betting is guaranteed, but it does provide a measurable benchmark against which model performance can be evaluated.

From a learning perspective, this project offers an opportunity to apply theoretical knowledge gained during the programme to a real-world problem. It requires skills in data collection, data cleaning, exploratory analysis, statistical modelling, and machine learning. These are all core competencies expected of a data science graduate. The football context makes the technical work more engaging and provides concrete examples that aid understanding.

Cognitive Biases in Betting Markets

An additional motivation for this research lies in understanding how cognitive biases affect betting behaviour and market efficiency. Behavioural economics has demonstrated that humans are not perfectly rational actors. Instead, decision-making is influenced by systematic biases that can be exploited or mitigated through better information systems.

The **recency bias** causes bettors to overweight recent results when forming expectations. A team that has won three consecutive matches may be perceived as stronger than their underlying quality suggests. Similarly, the **availability heuristic** leads people to overestimate the probability of memorable events such as shock cup upsets or high-profile collapses. These biases can create systematic mispricings in betting markets.

The **favourite-longshot bias** is well documented in sports betting literature. Bettors tend to overbet on longshots (high odds outcomes) and underbet on favourites. This is partly explained by **prospect theory**, which suggests that people overweight small probabilities and are attracted to the possibility of large gains even when expected value is negative. Understanding these biases is essential for developing models that can identify genuine value.

The **confirmation bias** affects how bettors interpret information. Fans of a particular team may selectively attend to positive news about their team while discounting negative indicators. This emotional attachment can lead to systematically biased probability assessments. A data-driven model, by contrast, processes all available information without emotional interference.

Figure 7: Cognitive Biases in Football Betting

COGNITIVE BIASES	ECONOMIC THEORIES
• Recency Bias Overweighting recent results	**• Prospect Theory** Loss aversion shapes decisions
• Availability Heuristic Memorable events seem likely	**• Expected Value** Probability × Payout analysis
• Confirmation Bias Seeking supporting evidence	**• Market Efficiency** Information reflected in odds
• Gambler's Fallacy Patterns in random events	**• Favourite-Longshot Bias** Longshots systematically overbet

↓ DATA SCIENCE APPROACH ↓

Systematic analysis removes emotional bias • Exploits market inefficiencies created by cognitive errors

(Understanding these biases informs both model design and ethical considerations)

Aim and Objectives

Research Aim

The aim of this research is to design and develop a football match outcome prediction model using statistical inference and machine learning techniques applied to data from Europe's top five leagues, and to evaluate its potential for identifying value betting opportunities.

Research Objectives

The following objectives have been formulated to achieve the research aim:

1. **To learn and understand** the fundamental statistical concepts underlying football match prediction, including Poisson distribution for goal modelling, expected goals (xG) metrics, and probability theory as applied to betting markets.
2. **To learn and understand** the application of machine learning classification algorithms such as logistic regression, random forest, and gradient boosting to sports outcome prediction problems.
3. **To design and develop** a prototype betting model that combines statistical inference with machine learning predictions to estimate match outcome probabilities and identify potential value bets.
4. **To obtain feedback and submit** the final thesis incorporating reflections on the learning process, model limitations, and ethical considerations surrounding predictive betting systems.

Table 1: Research Objectives Mapping

Objective	Description	Type
1	Statistical foundations of prediction	Learn and Understand
2	Machine learning for classification	Learn and Understand
3	Prototype betting model development	Design and Develop

Objective	Description	Type
4	Final thesis with reflections	Feedback and Submit

Justification

This research is justified on several grounds. Firstly, sports analytics is a rapidly growing field with significant industry demand. Major football clubs now employ data analysts and scientists to inform decisions on player recruitment, tactical planning, and opponent analysis. By developing skills in this area, the research contributes to career readiness in a competitive job market.

Secondly, football betting prediction represents a challenging classification problem that requires integration of multiple data science techniques. Unlike many academic datasets, football data is messy, incomplete, and subject to numerous confounding factors such as injuries, weather, and referee decisions. Working with such data develops practical skills that cannot be gained from sanitised textbook examples.

Thirdly, the project addresses the ethical dimensions of predictive modelling. Betting systems can cause harm if used irresponsibly. By explicitly considering these ethical issues as part of the research, the project demonstrates awareness of the broader societal implications of data science work. This aligns with the module's emphasis on understanding the impact of our work on others.

Finally, Europe's top five leagues provide a rich and diverse dataset. Each league has distinct characteristics in terms of playing style, competitive balance, and historical patterns. Analysing data across multiple leagues enables comparison and helps identify which findings are generalisable versus league-specific. This adds academic rigour to the research.

From a behavioural economics perspective, betting markets represent a natural laboratory for studying decision-making under uncertainty. The principles of **bounded rationality**—the idea that human cognitive limitations constrain optimal decision-making—are directly observable in betting behaviour. By developing models that account for these limitations, the research contributes to broader understanding of how data science can support better human decision-making while remaining mindful of the potential for harm.

Research Questions and Hypotheses

Research Questions

The following research questions will guide the investigation:

1. **RQ1:** To what extent can historical match statistics and derived features be used to predict football match outcomes across Europe's top five leagues?
2. **RQ2:** What are the ethical implications of developing and deploying football betting prediction models, and how can potential harms to vulnerable individuals be identified and mitigated?

Hypotheses

Based on preliminary literature review, the following hypotheses are proposed:

1. **H1:** Machine learning models incorporating form-based features, expected goals metrics, and team strength ratings will achieve prediction accuracy exceeding 50% for match outcomes (home win, draw, away win), which is higher than random baseline performance.
2. **H2:** The development of transparent prediction models with explicit probability outputs and uncertainty measures will enable more informed decision-making compared to opaque betting systems, thereby reducing the potential for harm associated with overconfident gambling behaviour.

Research Methodology

Desk-Based Agile Research Approach

This research will adopt a desk-based agile methodology. The desk-based nature reflects that all data collection and analysis will be conducted using secondary sources available through online databases and APIs. No primary data collection involving human participants is required, which simplifies the ethical approval process.

The agile approach is borrowed from software development practices but adapted for research purposes. Rather than following a rigid waterfall model where each phase must be completed before the next begins, agile methodology allows for iterative development. Work will be organised into sprints of approximately two weeks each. At the end of each sprint, progress will be reviewed, and the direction of subsequent work adjusted based on findings.

This approach is particularly suitable for data science projects where initial assumptions may prove incorrect once data is examined. For example, certain features might be unavailable for some leagues, or particular algorithms might prove computationally infeasible. The agile framework provides flexibility to adapt while maintaining overall progress toward objectives.

Data Collection Strategy

Historical match data will be collected from publicly available sources including Football-Data.co.uk, which provides comprehensive match results and betting odds for major European leagues dating back multiple seasons. Additional data on expected goals and advanced statistics may be obtained from sources such as FBref and Understat.

The dataset will cover an extensive period from 1990 to 2026 across all five target leagues. This provides approximately 70,000 matches spanning over three decades for analysis. This longitudinal dataset enables examination of how football has evolved over time and whether prediction models trained on historical data remain valid for modern football. Data cleaning will address missing values, inconsistent team names, and duplicate records. Feature engineering will create derived variables such as rolling averages, goal difference, and points per game.

Analysis Framework

Analysis will proceed in three phases. The first phase involves exploratory data analysis to understand distributions, correlations, and potential predictive features. The second phase applies statistical modelling, beginning with Poisson regression for goal prediction and Dixon-Coles adjustment for home advantage. The third phase implements machine learning classifiers and compares their performance against statistical baselines.

Model validation will use time-series cross-validation to respect the temporal nature of football data. Models will be trained on historical data and tested on subsequent seasons to simulate realistic prediction scenarios. Performance metrics will include accuracy, log-loss, and calibration curves comparing predicted probabilities to observed frequencies.

Figure 3: Agile Research Methodology Framework

Sprint 1-2: Data Collection & Cleaning
↓
Sprint 3-4: Exploratory Analysis & Feature Engineering
↓
Sprint 5-6: Statistical Modelling (Poisson/Dixon-Coles)
↓
Sprint 7-8: Machine Learning Implementation
↓
Sprint 9-10: Value Betting Analysis & Evaluation
↓
Sprint 11-12: Thesis Writing & Reflection

Literature Review

Statistical Approaches to Football Prediction

The academic study of football prediction has a long history. Maher (1982) was among the first to propose using Poisson regression to model goal-scoring in football matches. His approach treated goals scored by each team as independent Poisson processes with rates determined by team attacking and defending strengths. While this model captured basic dynamics, it failed to account for the correlation between home and away goals observed in actual matches.

Dixon and Coles (1997) addressed this limitation by introducing a correction factor for low-scoring matches. Their model remains influential and is still used as a baseline in contemporary research. The key insight was that matches with zero or one goal for each team occur more frequently than independent Poisson models predict. The Dixon-Coles adjustment accounts for this by modifying probabilities for scorelines like 0-0, 1-0, 0-1, and 1-1.

More recent statistical work has incorporated time-decay factors that give greater weight to recent results. Constantinou et al. (2012) demonstrated that team strengths vary over time and that models accounting for this temporal variation outperform static approaches. Their work also highlighted the importance of considering contextual factors such as match importance and fixture congestion.

Machine Learning in Sports Prediction

Machine learning approaches to football prediction have gained traction in recent years. Huang and Chang (2010) applied neural networks to predict English Premier League results and reported accuracy rates around 60%. However, their work has been criticised for potential overfitting and lack of proper temporal validation.

Tax and Joustra (2015) conducted a systematic comparison of machine learning methods on Dutch Eredivisie data. They found that random forest classifiers slightly outperformed logistic regression and naive Bayes approaches. Importantly, they emphasised the difficulty of beating bookmaker predictions, noting that even well-calibrated models struggled to achieve consistent profitability.

Hubacek et al. (2019) combined gradient boosting machines with extensive feature engineering including expected goals metrics. Their model achieved competitive performance on Kaggle competitions but the authors acknowledged that real-world deployment faces additional challenges including transaction costs and betting limits imposed by bookmakers.

Economic Theories and Market Efficiency

The **Efficient Market Hypothesis (EMH)** provides a theoretical framework for understanding betting markets. In its strong form, EMH suggests that all available information is immediately reflected in prices, making it impossible to consistently achieve above-market returns. Applied to betting, this would imply that bookmaker odds perfectly reflect true probabilities.

However, research suggests that betting markets exhibit **weak-form efficiency** at best. Kuypers (2000) found evidence of exploitable inefficiencies in English football betting markets, particularly around draws and away wins. Forrest and Simmons (2008) demonstrated that incorporating additional information sources could improve upon bookmaker predictions, suggesting markets do not fully incorporate all available data.

The concept of **expected value** from decision theory is central to this research. A bet has positive expected value when the probability of winning multiplied by the potential payout exceeds the stake. Identifying such opportunities requires accurate probability estimation, which is precisely what this project attempts to develop.

Game theory also informs our understanding of betting markets. Bookmakers and sophisticated bettors engage in strategic interaction where each party attempts to anticipate the other's behaviour. Bookmakers adjust odds based on betting patterns, while sharp bettors attempt to place wagers before odds move. This dynamic creates a complex adaptive system where simple prediction models may be insufficient.

Case Studies

Case Study 1 - FiveThirtyEight Soccer Power Index: The American media organisation FiveThirtyEight publishes match predictions based on their Soccer Power Index (SPI). The methodology combines adjusted goals, shot-based expected goals, and non-shot expected goals to rate team quality. The model is transparent about its predictions and provides probability estimates for each match outcome. Analysis of their historical predictions suggests reasonable calibration but limited ability to consistently beat closing betting odds.

Case Study 2 - Stratagem and Betting Syndicates: Professional betting syndicates reportedly use sophisticated models that incorporate not only match statistics but also market information such as odds movements and liquidity. While specific methodologies are proprietary, interviews and legal cases have revealed that successful operations typically combine statistical models with human expertise and sophisticated risk management. This suggests that purely automated approaches may be insufficient for consistent profitability.

Table 2: Literature Summary

Author(s)	Approach	Key Contribution
Maher (1982)	Poisson Regression	Foundation for goal-based modelling
Dixon & Coles (1997)	Adjusted Poisson	Low-score correlation correction

Author(s)	Approach	Key Contribution
Kuypers (2000)	Market Efficiency	Exploitable inefficiencies in betting markets
Kahneman (2011)	Behavioural Economics	Cognitive biases affecting decisions
Tax & Joustra (2015)	ML Comparison	Random forest superiority for classification
Hubacek et al. (2019)	Gradient Boosting + xG	Feature engineering with expected goals

Integration of Tools, Technologies and Techniques

The successful completion of this project requires integration of various tools, technologies and analytical techniques. The selection of these components has been guided by considerations of suitability, availability, and learning value.

Programming and Analysis Environment

Python will serve as the primary programming language for this research. Python offers extensive libraries for data manipulation, statistical analysis, and machine learning. The core libraries to be used include Pandas for data handling, NumPy for numerical computation, Scikit-learn for machine learning algorithms, and Statsmodels for statistical modelling. Visualisation will be conducted using Matplotlib and Seaborn.

Jupyter Notebooks will provide an interactive environment for exploratory analysis and documentation. This format allows code, output, and explanatory text to be combined in a single document, facilitating both analysis and communication of findings. For production code, standard Python scripts will be used with appropriate version control through Git.

Statistical and Machine Learning Techniques

The statistical component of the model will employ Poisson regression with Dixon-Coles adjustment. This technique treats goal-scoring as a Poisson process where the expected goals for each team depend on attack and defence parameters. The implementation will follow published methodologies while adapting for the specific dataset characteristics.

Machine learning classification will utilise logistic regression as a baseline, followed by ensemble methods including random forest and XGBoost. Hyperparameter tuning will be conducted using grid search with cross-validation. The choice of these algorithms reflects the balance between interpretability and predictive power appropriate for this research context.

Table 3: Technology Stack

Category	Tool/Technology	Purpose
Programming	Python 3.x	Primary development language
Data Handling	Pandas, NumPy	Data manipulation and computation
Machine Learning	Scikit-learn, XGBoost	Classification algorithms
Statistics	Statsmodels, SciPy	Statistical modelling and inference
Visualisation	Matplotlib, Seaborn	Charts and graphical output
Data Sources	Football-Data.co.uk, FBref	Historical match and betting data

Figure 4: Data Pipeline Architecture

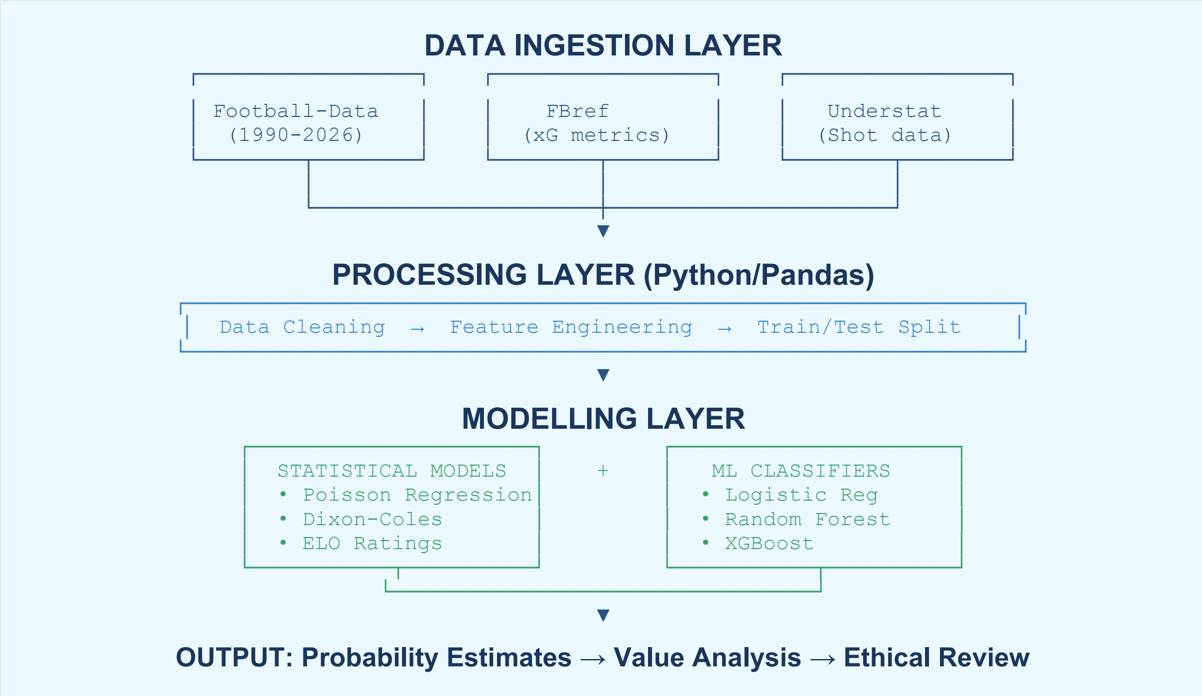

Figure 5: Feature Engineering Framework

RAW FEATURES	DERIVED FEATURES
• Match results (H/D/A) • Goals scored/conceded • Shots on target • Corners and fouls • Bookmaker odds • Date and venue	• Rolling form (5/10 games) • Goal difference trend • Expected goals (xG) • ELO/Power ratings • Head-to-head history • Home advantage factor

Figure 6: Model Comparison Framework

BASELINE	STATISTICAL	MACHINE LEARNING
Random Guess	Poisson/Dixon-Coles	RF/XGBoost
~33% Accuracy	*~45-50% Accuracy*	*~50-55% Accuracy*
Benchmark	Interpretable	Higher Complexity

(Expected accuracy ranges based on literature review)

Conclusion

This proposal has outlined a research project focused on developing a football betting prediction model using data from Europe's top five leagues. The project combines statistical inference techniques with machine learning approaches to address the challenging problem of match outcome prediction. The primary contribution of this research will not be a commercially viable betting system. Rather, it will be a deep understanding of the complexities involved in sports prediction and the practical application of data science techniques to real-world problems. The process of collecting, cleaning, and analysing football data will develop skills that are transferable to other domains. The ethical dimensions of betting prediction have been acknowledged throughout this proposal. While the research does not directly encourage gambling, it recognises that prediction models could be misused. The final thesis will include explicit discussion of responsible gambling and the limitations of predictive models.

In summary, this project represents an opportunity to apply theoretical knowledge to an engaging practical problem while developing the critical thinking and reflection skills expected of a masters-level graduate. The emphasis on learning rather than perfection aligns with the module's educational objectives and will produce a thesis that demonstrates genuine understanding rather than superficial technical achievement.

References

[1] Constantinou, A.C., Fenton, N.E. and Neil, M. (2012) 'pi-football: A Bayesian network model for forecasting Association Football match outcomes', *Knowledge-Based Systems*, 36, pp. 322-339.

[2] Dixon, M.J. and Coles, S.G. (1997) 'Modelling association football scores and inefficiencies in the football betting market', *Journal of the Royal Statistical Society: Series C*, 46(2), pp. 265-280.

[3] Forrest, D. and Simmons, R. (2008) 'Sentiment in the betting market on Spanish football', *Applied Economics*, 40(1), pp. 119-126.

[4] Huang, K.Y. and Chang, W.L. (2010) 'A neural network method for prediction of 2006 World Cup Football Game', *Proceedings of the International Joint Conference on Neural Networks*, pp. 1-8.

[5] Hubacek, O., Sourek, G. and Zelezny, F. (2019) 'Exploiting sports-betting market using machine learning', *International Journal of Forecasting*, 35(2), pp. 783-796.

[6] Kahneman, D. (2011) *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux.

[7] Kahneman, D. and Tversky, A. (1979) 'Prospect Theory: An Analysis of Decision under Risk', *Econometrica*, 47(2), pp. 263-291.

[8] Kuypers, T. (2000) 'Information and efficiency: an empirical study of a fixed odds betting market', *Applied Economics*, 32(11), pp. 1353-1363.

[9] Maher, M.J. (1982) 'Modelling association football scores', *Statistica Neerlandica*, 36(3), pp. 109-118.

[10] Tax, N. and Joustra, Y. (2015) 'Predicting the Dutch football competition using public data: A machine learning approach', *Transactions on Knowledge and Data Engineering*, 10(10), pp. 1-13.

[11] Thaler, R.H. and Sunstein, C.R. (2008) *Nudge: Improving Decisions about Health, Wealth, and Happiness*. New Haven: Yale University Press.

[12] Tversky, A. and Kahneman, D. (1974) 'Judgment under Uncertainty: Heuristics and Biases', *Science*, 185(4157), pp. 1124-1131.